

PREDICATES AND NESTED QUANTIFIER

- ❖ Quantifiers
- ❖ Translating English Sentences into Logical Expressions

Learning Objective

- ❖ To understand the types of quantifiers

Quantifiers

❖ Universal Quantification

- A predicate is true for every element under consideration.

❖ Existential Quantification

- There is one or more element under consideration for which the predicate is true.

- ❖ The area of logic that deals with predicates and quantifiers is called the *predicate calculus*.

Universal Quantifier($\forall$)

- ❖ The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$
- ❖ Read as “for all $x P(x)$ ” or “for every $x P(x)$ ”.
- ❖ “all of ”, “for each ”, “given any ”, “for arbitrary ”, “for each”
- ❖ $\forall x P(x)$: $P(x)$ for all values of x in the domain.

Universal Quantifier(Examples)

- ❖ Let $P(x)$ be the statement " $x + 1 > x$." $\forall x P(x)$ is true.
- ❖ Let $Q(x)$ be the statement " $x < 2$." $\forall x Q(x)$ is false.
- ❖ $P(x)$ is " $x^2 > 0$." $\forall x P(x)$ is false.
- ❖ $\forall x (x^2 \geq x)$ if the domain consists of all real numbers is false.
- ❖ $\forall x (x^2 \geq x)$ if the domain consists of all integers is true.

Existential Quantifier($\exists$)

- ❖ The notation is $\exists x P(x)$.
- ❖ “There exists an element x in the domain such that $P(x)$.”
- ❖ Read as “There is an x such that $P(x)$,”
“There is at least one x such that $P(x)$,”
or For some $x P(x)$.

Existential Quantifier($\exists$)

❖ Let $P(x)$ denote the statement “ $x > 3$.”

$\exists xP(x)$ is true because $P(4)$ is true.

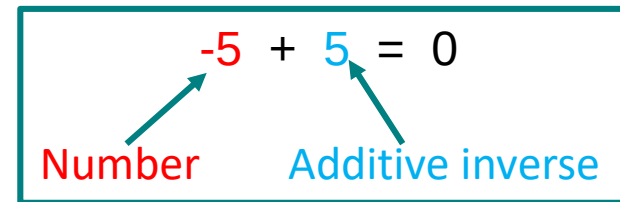
❖ Let $Q(x)$ denote the statement “ $x = x+1$.”

$\exists xQ(x)$ is false because $Q(x)$ is false for all real number x .

Nested Quantifier

- ❖ The nested quantifiers, where one quantifier is within the scope of another, such as
$$\forall x \exists y (x + y = 0).$$
- ❖ For every real number x , there is a real number y such that $x + y = 0$.

- ❖ Every real number has an additive inverse


$$\text{Number} \rightarrow -5 + 5 = 0 \leftarrow \text{Additive inverse}$$

Nested Quantifier

- ❖ The statement $\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0))$ says that for every real number x and for every real number y , if $x > 0$ and $y < 0$, then $xy < 0$.
- ❖ **Translate into English:** “The product of a positive real number and a negative real number is always a negative real number.”

Summary

- ❖ Types of quantifiers (**Universal quantifier and Existential quantifier**)
- ❖ Nested Quantifiers



Let $P(x)$ be the statement " $x + 1 > x$." What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?

Answer: TRUE



Next Lecture



Sets and Set Operations